what lame super power would you rather have?

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Q1

~Code~
340

CS 340

Concurrency | OS

Today's art

Updates

0.5%

1. MP5 Allocator due Tuesday

2. 2nd Chance Exam 1 Grades Released
   a. Regrade requests by next week Thursday
   b. Submit an issue on Prairie Learn

3. Exam 2 sign up open

4. HW 5 out now, due next Wednesday

Concurrency | OS

Today's LGs - Expand your mental model of a computer to include running multiple program concurrently.

Be able to answer conceptual questions about concurrency and the OS

Agenda

1. Review
2. Concurrency with Processes
3. Operating System (OS)
4. Practice Clickers

What is true about a CPU?

T A) It does fetch, decode, and execute cycles

T B) It can decode and execute basic instructions like jump, load, store, math

F ~~C)~~ It can decode and execute C code directly

T D) It uses gates to implement logic

F ~~E)~~ It ~~*optionally*~~ has a program counter register

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Q2

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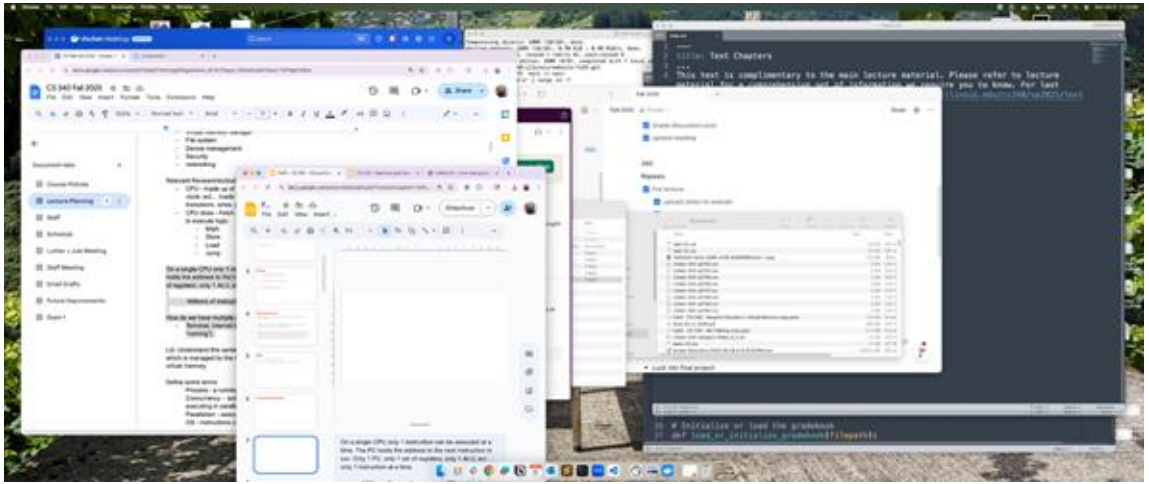
How?

a single CPU

- 1 instruction at a time



Only 1 PC

1 set of registers

1 ALU

LG - Be able to explain each part of this sentence

Processes are running concurrently which is managed by the OS and is safe because of kernel mode and virtual memory.

Vocabulary

Process - a running instance of a program

Concurrency - active during the same time

Parallelism - executing at the same time

At 5:15pm how many activities am I doing concurrently? In parallel?

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Q3

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I start my stats HW at 4:00pm, at 5:00pm I take a break start a game of volleyball which ends at 6:00pm, I then work on and finish my stats HW at 8:30pm.

con: 2
↑
concurrently

Par: 1
↑
parallel

At 7:00pm how many activities am I doing concurrently? In parallel?

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Q4

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I start my stats HW at 4:00pm, at 5:00pm I take a break start a game of volleyball which ends at 6:00pm, I then work on and finish my stats HW at 8:30pm.

con: 1
par: 1

LG - Be able to explain each part of this sentence

Processes are running concurrently which is managed by the OS and is safe because of kernel mode and virtual memory.

Processes are running concurrently

page table / VM

CPU state → registers
PC

unique process ID

What is the closest estimate to how many instructions a CPU can run in a second?

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Q5

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1,000,000

If I'm typing at 200 characters a minute, how many CPUs do I need to seamlessly handle updating my notes app while running my slack app (chat) to monitor if I got any new messages?

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Q6

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1 one

The Operating System - Kernel

- Process management
  - Create, destroy, schedules

Info
- Page table
- PC
- registers
- ID

context switching
- stores info ~~loads~~
- Load info new process

- File system - reading/writing, permissions
- Devices (keyboard)
- Security - accounts
- Networking

Processes are running concurrently which is managed by the OS

- schedules and loads processes
  - every few milliseconds there is a timer interrupt
    - potentially context switches
    - saves info PCB (process control block)

      every active process has a PCB
- system call

How do user programs interact with the OS?

User Mode - code you wrote

Kernel Mode - only kernel part of the OS is running

- change something another process could notice
- Observe something another process could have done
- Observe something the user did
- claim or releasing system resources

Kernel Mode

system call

- switches to kernel code
- does the thing you requested
- put something back in the CPU

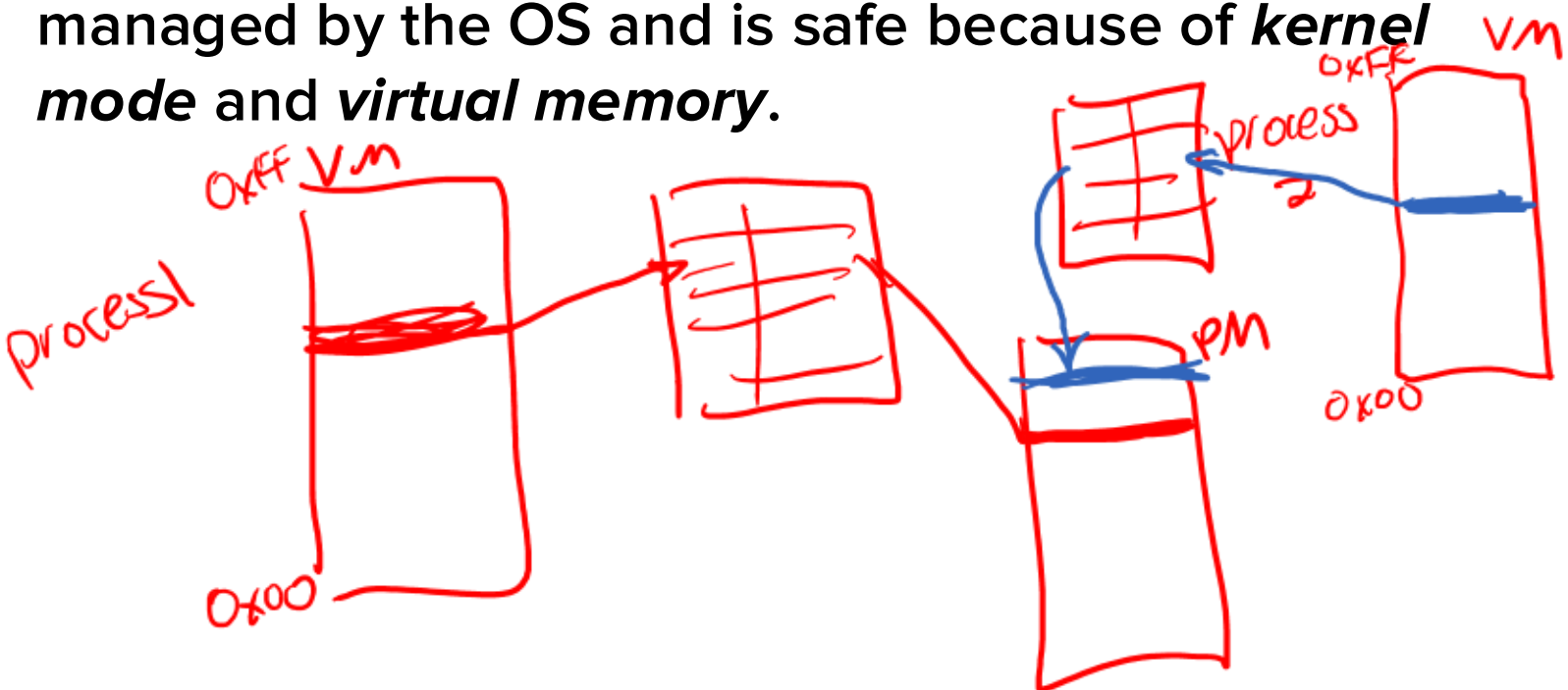
PBC

fopen() Example

- instruction trigger a sys call
  - switches to kernel mode
- load file into memory from disk
- OS context switches to my stardew valley
- CPU get an interrupt from disk
- Context switch back

LG - Be able to explain each part of this sentence

Processes are running concurrently which is managed by the OS and is safe because of ***kernel mode*** and ***virtual memory***.



Is it possible for these two programs to print out the same value while running concurrently?

PBC

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Q7

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```
int x = 5;
printf("%#x", &x);
```

virtual memory

```
int x = 2;
printf("%#x", &x);
```

yes

Is it possible for these two programs to be using the same register ID to store the value x for calculations if they are running concurrently?

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Q8

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CPU

~ str 5 regB

~ str 2 regB

```
int x = 5;

x = x + 1;
```

```
int x = 2;

x = x + 1;
```

yes

What does the OS not do?

A) Schedule Processes
B) Execute all instructions
C) Manage virtual memory
D) Allow the keyboard to talk to your program
~~E) Does all of the above~~

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Q9

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What manages the concurrent processes on your computer?

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Q10

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A) CPU
B) OS
C) ALU
D) IDE

Can a process search through and edit anything on your computer? Why or why not?

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Q11

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NO

kernel
VM

How does your C code interact with the OS?

- system calls

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Q12

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How do the following connect or relate to each other?

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Q13

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Transistors, gates, mux, cpu, processes, kernel mode, virtual memory, OS, c code

LG - Be able to explain each part of this sentence

Processes are running concurrently which is managed by the OS and is safe because of kernel mode and virtual memory.